

RICH TILLMAN

Principal Frontend Engineer

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SUMMARY

Principal Frontend Engineer, 15 years - founding/architecture-level engineer across five 0-to-1 startups, IC to Engineering Director, with real experience running teams, not just shipping UI. The throughline across every role: Nx monorepo architecture, structural and lint tooling (now Oxlint), and the design-system to shared-UI-library to frontend to BFF to backend pattern, built repeatedly at Freebird, Pineapple, PHC Global, and Redesign Health. Early, genuine adopter of React and Storybook (2016 - still active in both communities' Discords), and pioneered browser-based AR for NARS Cosmetics years before WebAR went mainstream. Deeply agent-native: creator of ForgeKit, an open-source CLI + MCP server suite (6,200+ npm installs across three separate first-time package releases) that is itself harness/spec tooling for AI coding agents - I design what context an agent needs to do a task right, not just prompt it. Claude Code and Cursor daily, including production use of Cursor's background agent. Seeking Principal / Staff Frontend, Product Engineer, or engineering-leadership roles where architecture, AI-native tooling, and real ownership meet.

EXPERIENCE

Consultant, AI Model Evaluation & Red-Teaming - Independent (Mercor, micro1, Handshake)

Feb 2026 - Jul 2026

- Evaluated, ranked, and red-teamed frontier AI model outputs against scoring rubrics on React/TypeScript frontend-engineering tasks across three sequential contract engagements (Mercor, then micro1, then Handshake); authored adversarial test cases to expose model failure modes and break outputs. Worked in ChatGPT Pro and Cursor across four frontier LLMs throughout.

Staff Frontend Engineer & Tech Lead to Engineering Director - Redesign Health

Jul 2022 - Jan 2026

- Promoted twice to Engineering Director; led delivery across three cross-functional teams (15+ members) while staying hands-on as a frontend architecture SME.
- Spearheaded a React design system of 50+ reusable components (Storybook + Chromatic) adopted across teams: 30% dev-time reduction across a 10-15 engineer org.
- Engineered 20+ data-visualization components with custom hooks and Zustand: 40% rendering-overhead reduction on large datasets; established a Chromatic visual-regression workflow.
- Adopted Ponicode for AI-generated React unit-test scaffolding, raising test coverage on new components with less manual boilerplate.
- Built an onboarding methodology that trained backend engineers to production-level React in one month.
- Drove build-tooling modernization: migrated to Vite (became the org's default bundler by early 2023) and to Tailwind CSS for an internal sister app to the client portal; migrated the team's default e2e suite from Cypress to Playwright; generated type-safe React Query hooks from OpenAPI/Swagger specs; collaborated with backend on JWT auth and authored the specs for an OIDC migration that was shelved when the team was restructured.

Senior Frontend Engineer & Tech Lead - Pineapple Corporation

Jan 2022 - Jul 2022

- Architected Nx monorepo with a design system based shared UI library, a cross-platform (iOS, Android, web) application, and a NestJS backend.
- Built cross-platform architecture with Expo, Nx, and React Native (25% mobile performance gain for 100K+ users).

Founding Frontend Engineer - PHC Global

Jul 2021 - Jan 2022

- Founding frontend architect for a healthtech/biosecurity startup: built the core product for enterprise clients (including Facebook and Amex, our first two customers) - a Mapbox GL heatmap of active COVID risk zones that clients used to schedule ground teams, plus flight/travel-arrangement suggestions and a separate interactive map for simulating alternate entry routes.
- Set up the foundational architecture: Nx monorepo with 30+ shared libraries and gRPC middleware (40% developer-experience improvement); lifted NPS-measured satisfaction 35%; cut infrastructure costs 30% via GCP, gRPC, and Kubernetes.
- Standardized client state with Redux Toolkit (RTK) and server state with React Query across the platform.

Lead Frontend Engineer - Freebird

Sep 2016 - Jan 2021

- Built one of the earliest production React UI libraries on Storybook: a 200-component design system spanning B2B, B2C, and React Native, eliminating the need for separate iOS and Android teams.
- Migrated the team's Redux + Redux-Saga boilerplate to Redux Toolkit (RTK) in 2019, cutting state-management boilerplate and standardizing patterns across the design system.
- Served as design-engineering liaison across client, sales, and marketing; shipped B2B/B2C and internal dashboards (React, NestJS, D3.js) and a React Native app with real-time data views.

Lead Web Developer - FaceCake Marketing Technologies (Los Angeles, CA)

Oct 2010 - Sep 2016

- Pioneered browser-based AR try-on for NARS, one of the first of its kind: compiled OpenCV to JavaScript via Emscripten for real-time client-side computer vision over getUserMedia, with Pixi.js, WebGL, and Three.js powering real-time face-tracking and makeup-rendering. Worked directly with Pixi.js's creator on techniques no one had attempted before, years ahead of mainstream WebAR.
- Shipped the NARS virtual try-on web app (~\$400K/month revenue lift) backed by a real-time CMS on Firebase Realtime Database; owned the full lifecycle end-to-end.

PROJECTS

ForgeKit - independent open-source developer-tooling venture (2024 - 2026) - Open-source CLI + MCP server suite bridging Figma to React to Storybook for design-system teams - itself a harness/spec-tooling product for AI coding agents (Claude Code, Cursor). 6,200+ npm installs across three separate first-time package releases, with active production adoption. forgekit.cloud, github.com/effinrich, npmjs.com.

ForgeKit Core CLI (forgekit.cloud) - TypeScript CLI scaffolding production-ready Nx monorepos with React 18, Storybook 10+, Vitest, Playwright, and GitHub Actions CI/CD; targets Chakra UI, shadcn/ui, and Tamagui across web and universal React Native. Used recursively to scaffold ForgeKit itself.

ForgeKit Figma MCP (npmjs.com) - MCP server extracting Figma variables and design tokens; generates typed theme configs for Chakra UI, Tailwind, and shadcn - enabling AI-driven design-to-code workflows.

ForgeKit Storybook MCP (npmjs.com) - MCP server exposing Storybook metadata, argTypes, and usage patterns to AI coding agents - automating story generation, docs scaffolding, and component-testing workflows. Direct, hands-on experience designing what context an AI agent needs to do a task correctly.

Perch - open-source voice-assistant widget - Drop-in voice AI for any website (one script tag + React SDK) built on ElevenLabs Conversational AI: dependency-free Shadow-DOM core, React SDK, and a serverless signed-URL proxy that keeps the API key server-side. Built end-to-end with Claude Code.

Santa Chat AI - AI-powered Santa Claus experience - Real-time chat, live phone calls, and SMS with Santa via ElevenLabs text-to-speech and Twilio, an LLM chat backend, and Stable Diffusion image generation; Stripe-based subscription tokens. Per-user personalization (parents added their child's name so Santa could address them by name) and analytics instrumentation (Amplitude, Google Analytics 4) tracked engagement in production. Next.js frontend/BFF layer proxying to a dedicated backend service. Selected by ElevenLabs for a 300K-token grant. Next.js, React, Chakra UI, Zustand, MongoDB, NextAuth.

Textation - AI character chat and avatar platform - Pivoted from Santa Chat AI into a general AI-companion product: custom AI characters with LLM chat, ElevenLabs text-to-speech, Deepgram speech-to-text, image

generation, SadTalker talking-head video, and live WebSocket voice chat; Stripe payments, Twilio SMS/voice. Next.js, React, Chakra UI, Zustand, MongoDB.

MCP Atlas (mcp-atlas-orcin.vercel.app) - Curated, link-verified directory of Model Context Protocol servers (18 official + community, searchable and filterable). Built with Claude Code. Astro (Zod-validated content collections + islands architecture), TypeScript, Vitest, and Playwright; Lighthouse 100 accessibility, 97 performance.

TokenCast (tokencast.vercel.app) - Paste design tokens (Figma variables, CSS custom properties, or a Tailwind config) and get a live preview plus exportable theme code for Tailwind, Chakra, and shadcn/ui, with server-rendered Save & Share links. Built with Cursor. React Router (framework mode: SSR, loaders, actions), React 19, TypeScript, Tailwind CSS 4, Supabase (RLS + SECURITY DEFINER RPCs, IP rate-limiting), Vitest + Playwright; Lighthouse 100 accessibility, 99 performance.

Tidy App (tidyapp.me) - Offline-first React Native + Expo app; Zustand + TanStack Query; Supabase; Figma Code Connect; 90% test coverage. Built with both Claude Code and Cursor, including production use of Cursor's background agent (verified via git co-author trailers on merged PRs).

Serial One (app.serialone.io) - Dec 2021 - Jan 2022 - Turborepo monorepo browsing live Candy Digital NFT sale listings (MLB-licensed digital trading cards), pulling data from MongoDB and filterable by player name and rarity tier, rendered in a masonry grid (price, sale date, edition number per card). Coinbase Commerce crypto-donation checkout integrated in the nav. Next.js, React, Chakra UI, MongoDB, axios, SWR.

TECHNICAL SKILLS

Languages: JavaScript (15 yrs), TypeScript (strict), HTML5, CSS3, SQL, PHP (legacy)

Frameworks & Libraries: React 19, Next.js, React Native, Expo, TanStack Start, Node.js / NestJS, React Hooks, Concurrent Rendering

Graphics & Real-Time Rendering: Pixi.js, WebGL, Three.js, Emscripten (C++ to JS/WASM), OpenCV, real-time client-side computer vision

Design Systems & Tools: Storybook 10+, Chromatic, Chakra UI, shadcn/ui, Tamagui, Ark UI, Radix UI, Tailwind CSS, Styled Components / CSS-in-JS, Figma Code Connect

Agentic AI Development: Model Context Protocol (MCP) - author production MCP servers, not just a consumer; Claude Code and Cursor (including Cursor's background/autonomous agent) in daily production use; ChatGPT Pro; Ponicode (AI-generated unit tests); Figma MCP; AI model evaluation & red-teaming across frontier LLMs

DevOps & Monorepo: Nx, Turborepo, pnpm workspaces, Module Federation, Vite, Oxcint, GitHub Actions, EAS Build, GCP, Kubernetes, gRPC

State & Data: Redux Toolkit, Zustand, TanStack Query, REST, GraphQL, tRPC, OpenAPI/Swagger (client & hook codegen), JWT, Supabase, PostgreSQL, Context API

Testing & Accessibility: Vitest, Jest, Playwright, Cypress, React Testing Library, WCAG 2.1 AA